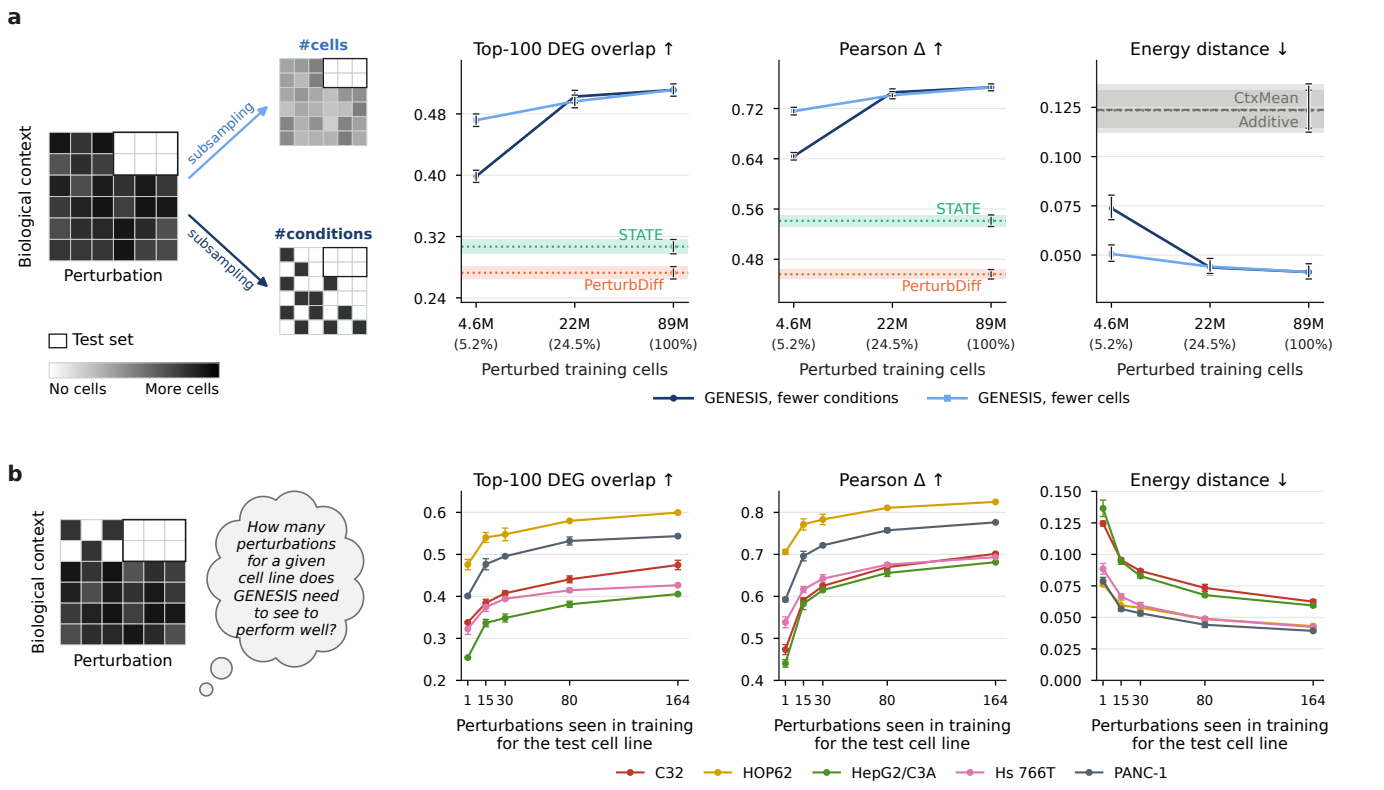

a
Biological context
Perturbation
subsampling
#cells
subsampling
#conditions
Test set
No cells
More cells
Top-100 DEG overlap ↑
0.48
0.40
0.32
0.24
STATE
PerturbDiff
4.6M (5.2%)
22M (24.5%)
89M (100%)
Perturbed training cells
Pearson Δ ↑
0.72
0.64
0.56
0.48
STATE
PerturbDiff
4.6M (5.2%)
22M (24.5%)
89M (100%)
Perturbed training cells
Energy distance ↓
0.125
0.100
0.075
0.050
CtxMean
Additive
4.6M (5.2%)
22M (24.5%)
89M (100%)
Perturbed training cells
GENESIS, fewer conditions
GENESIS, fewer cells
b
Biological context
Perturbation
How many perturbations for a given cell line does GENESIS need to see to perform well?
Top-100 DEG overlap ↑
0.6
0.5
0.4
0.3
0.2
1 15 30 80 164
Perturbations seen in training for the test cell line
Pearson Δ ↑
0.8
0.7
0.6
0.5
0.4
1 15 30 80 164
Perturbations seen in training for the test cell line
Energy distance ↓
0.150
0.125
0.100
0.075
0.050
0.025
0.000
1 15 30 80 164
Perturbations seen in training for the test cell line
C32
HOP62
HepG2/C3A
Hs 766T
PANC-1